Mauricio Gómez López

Curriculum Vitae

University of Southern Denmark
Department of Mathematics
Odense, Denmark
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Academic Positions

2025- **Assistant Professor**, *University of Southern Denmark*, Denmark.
Joint appointment at the Department of Mathematics and the STEM Education Research Center.

2020-2025 **Visiting Assistant Professor**, *Lafayette College*, United States.

2017-2020 **Paul Olum Postdoctoral Scholar**, *University of Oregon*, United States.

2015-2017 **Scientific Assistant**, *University of Southern Denmark*, Denmark.

Education

2011- 2015 **Ph.D. in Mathematics**, *University of Copenhagen*, Denmark.
Dissertation: *Spaces of piecewise linear manifolds.*

2009-2011 **M.S. in Mathematics**, *University of Copenhagen*, Denmark.

2005-2009 **B.S. in Mathematics**, *Universidad Nacional de Colombia*, Colombia.

Research Interests

Matroid Theory, Algebraic Topology, Mathematics Education.

Publications and Preprints

1. **The homotopy type of the topological cobordism category,** with Alexander Kupers. *Documenta Mathematica* 27, 2107-2182 (2022). arXiv:1810.0527.
2. **The homotopy type of the PL cobordism category. I.** *Algebraic and Geometric Topology.* Volume 25, Issue 5, 2527-2611 (2025). arXiv:1608.06236.
3. **The homotopy type of the PL cobordism category. II.** *Algebraic and Geometric Topology.* Volume 25, Issue 5, 2613-2665 (2025). arXiv:2308.04612.
4. **Realizing the Tutte polynomial as a cut-and-paste K-theoretic invariant.** To be submitted to *Homology, Homotopy and Applications.* arXiv:2501.12250.

Papers in Preparation

1. **The magnitude of a matroid.**
2. **Schema-based instructional design in pure mathematics.**

Supervised Undergraduate Research Projects

1. **Matroids from tilings.** Student: Nitan Avivi (University of Oregon, Fall 2019).
2. **Cohomology theory.** Student: Søren Eriksen (U. of Southern Denmark, Spring 2016).

Teaching Experience (as Instructor of Record)

University of Southern Denmark

Fall 2026 **Stochastic Differential Equations**
Fall 2026 **Ordinary Differential Equations**
Fall 2026 **Elementary Discrete Mathematics**
Spring 2026 **Mathematical and Numerical Analysis**
Fall 2025 **Functional Analysis**
Fall 2025 **Stochastic Differential Equations**

Lafayette College

Spring 2025 **Vector Spaces (Advanced Undergraduate Course)**
Spring 2025 **Statistical Concepts (Two Sections)**
Fall 2024 **Applied Statistics (Two Sections)**
Fall 2024 **Statistical Concepts**
Spring 2024 **Linear Algebra with Applications (Two Sections)**
Spring 2024 **Applied Statistics**
Fall 2023 **Modeling and Differential Calculus**
Fall 2023 **Calculus II (Two Sections)**
Spring 2023 **Modeling and Differential Calculus**
Spring 2023 **Calculus II (Two Sections)**
Fall 2022 **Calculus I (Two Sections)**
Fall 2022 **Complex Analysis (Advanced Undergraduate Course)**
Spring 2022 **Applied Statistics**
Spring 2022 **Calculus III (Two Sections)**
Fall 2021 **Calculus III**
Fall 2021 **Calculus I (Two Sections)**
Spring 2021 **Linear Algebra with Applications**
Spring 2021 **Calculus II (Two Sections)**
Fall 2020 **Calculus I (Three Sections)**

University of Oregon

Spring 2020 **Fundamentals of Analysis II**
Spring 2020 **Fundamentals of Analysis I**
Winter 2020 **Fundamentals of Analysis II**
Fall 2019 **Fundamentals of Analysis I**
Spring 2019 **Elements of Discrete Mathematics II**
Winter 2019 **Fundamentals of Analysis I**
Winter 2019 **Elementary Linear Algebra I**
Fall 2018 **Morse Theory (Graduate-Level Course)**
Spring 2018 **Introduction to Differential Equations**
Spring 2018 **Elements of Discrete Mathematics I**
Fall 2017 **Calculus I**

University of Southern Denmark

Spring 2017 **Generalized Linear Models**

Spring 2016 **Algebraic Topology (Undergraduate Reading Course)**

Spring 2016 **Algebra II: Introduction to Rings and Groups**

Spring 2016 **Vector Calculus**

Fall 2015 **Probability and Statistics**

Spring 2015 **Algebra II: Introduction to Rings and Groups**

Presentations

24 February 2020 **Matroid theory: A unified theory of independence,** *Undergraduate Mathematics Colloquium at The College of Wooster.* Wooster, Ohio.

8 October 2019 **Matroid theory: A unified theory of independence,** *Algebra and Combinatorics Seminar at the Claremont Center for the Mathematical Sciences.* Claremont, California.

8 October 2019 **Cobordism categories and spaces of manifolds,** *Topology Seminar at the Claremont Center for the Mathematical Sciences.* Claremont, California.

14 April 2018 **The homotopy type of the topological cobordism category,** *AMS Western Sectional Meeting.* Portland, Oregon.

21 November 2017 **The homotopy type of the PL cobordism category,** *Topology Seminar at the University of Oregon.* Eugene, Oregon.

7 September 2015 **The homotopy type of the PL cobordism category,** *Third Congress of Young Researchers of the Royal Spanish Mathematical Society.* Murcia, Spain.

6 June 2015 **Spaces of piecewise linear manifolds,** *Topology Seminar at the University of Münster.* Münster, Germany.

7 November 2014 **Spaces of piecewise linear manifolds,** *Topology Seminar at the University of Copenhagen.* Copenhagen, Denmark.

Awards and Nominations

2026 **Faculty of Science Award for Outstanding Teaching - University of Southern Denmark.** Nominated.

2022-2024 **McKelvy's Teaching Recognition.** Teaching recognition awarded by McKelvy scholars, Lafayette College's group of highest achieving students.

2024 **Aaron O. Hoff Teaching Award - Lafayette College.** Nominated.

2009 **Danish Governmental Scholarship.** Scholarship consisting of a tuition waiver and financial support to cover living expenses throughout my graduate studies.

2009 **Colfuturo Scholarship.** Merit-based scholarship granted to Colombian academics admitted to a graduate program abroad.

Outreach and Service

2020-2022 **Mathematics Colloquium**, Lafayette College (Organizer).

2019-2020 **McNair Scholars Program**, University of Oregon (Mentor).

Conferences Attended

January 2024 **Joint Mathematics Meetings 2024**, San Francisco, California.

January 2020 **Joint Mathematics Meetings 2020**, Denver, Colorado.

May 2019 **62nd Cascade Topology Seminar**, *Portland State University*, Portland, Oregon.

April 2018 **AMS Western Sectional Meeting**, *Portland State University*, Portland, Oregon.

June 2016 **Topology of Manifolds: A conference in honor of Michael Weiss's 60th birthday**, *University of Lisbon*, Lisbon, Portugal.

September 2015 **Third Congress of Young Researchers of the Royal Spanish Mathematical Society**, *University of Murcia*, Murcia, Spain.

July 2014 **Young Topologists Meeting 2014**, *University of Copenhagen*, Copenhagen, Denmark.

June 2014 **Manifolds, K-theory, and related topics**, *Inter-University Center*, Dubrovnik, Croatia.

July 2013 **Young Topologists Meeting 2013**, *EPFL*, Lausanne, Switzerland.

August 2013 **Masterclass on Homological Stability**, *University of Copenhagen*, Copenhagen, Denmark.

April 2013 **Masterclass on String Topology**, *University of Copenhagen*, Copenhagen, Denmark.

August 2012 **Fourth Arolla Conference on Algebraic Topology**, Arolla, Switzerland.

July 2012 **Young Topologists Meeting 2012**, *University of Copenhagen*, Copenhagen, Denmark.

June 2012 **Topology and Groups**, *Freie Universität Berlin*, Berlin,Germany.

Conferences Organized

July 2014 **Young Topologists Meeting 2014**, *University of Copenhagen*, Copenhagen, Denmark.

Professional Development

Spring 2024 **International Faculty Community of Practice**, Lafayette College.

Fall 2021 **Inclusive Instructors Academy**, *semester-long program on inclusive teaching strategies and methodologies*, Lafayette College.

2020-2021 **New Faculty Learning Community**, *year-long orientation for new faculty at Lafayette College*, Lafayette College.

July 2019 **Summer Institute on Scientific Teaching**, *week-long workshop focused on strategies for fostering diversity, equity, and inclusion in scientific teaching*, University of Oregon.

April 2019 **Engaging Students and Making Classrooms Fair and Inclusive: Cross-Disciplinary Strategies and Language to Promote Student Success**, *workshop led by Professor Kimberly Tanner*, University of Oregon.

Academic Visits

Spring 2013 (3 weeks) **University of Cambridge.** Cambridge, United Kingdom. Host: Oscar Randal-Williams.

Fall 2013 (6 weeks) **University of Cambridge.** Cambridge, United Kingdom. Host: Oscar Randal-Williams.

Spring 2014 (6 weeks) **University of Cambridge.** Cambridge, United Kingdom. Host: Oscar Randal-Williams.

Language Skills

Spanish Native

English Fluent

Norwegian Proficient

Danish Proficient

References

Daniel Dugger, University of Oregon ddugger@uoregon.edu

Martin Svensson, U. of Southern Denmark svensson@imada.sdu.dk

Louis Zulli, Lafayette College zullil@lafayette.edu